SHANYU CHAUHAN

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EDUCATION

University of California, San Diego (2024 – 2028) GPA: 3.8

- B.S. Mechanical Engineering (Robotics & Controls)

University of Oxford, United Kingdom GPA: 4.0

- Academic summer program on Artificial Intelligence & Machine Learning, NLP **(Jun 2025 – Aug 2025)**
- Research paper on 'Robot human interaction' submitted to MIT Journal **(Oct 2025 – Present)**

TECHNICAL SKILLS

- **CAD & Fabrication:** SolidWorks, Fusion 360, Onshape, OpenRocket, GD&T, 3D printing, CNC, lathe, soldering
- **Robotics & Controls:** Arduino, ESP32, circuit design and breadboarding, computer vision (OpenCV), MATLAB
- **Software & AI/ML:** Python, C++, PyTorch, TensorFlow, Pandas, deep neural networks, Git/GitHub, Azure, SSH, AI Agents

ENGINEERING PROJECTS

Drug Discovery Optimization Machine | Self Directed Research and Product Development **Jun 2026 - Present**

- Developing a single benchtop device that consolidates multiple sample preparation steps into a single device. Eliminates manual sample transfers. Opportunity identified through user discovery interviews with formulation scientists. *(Mechanism details withheld pending IP review.)*
- Projected to eliminate manual equipment swapping and save scientists 5+ hours per week of sample prep time
- Building the full system end to end across multiple design iterations from coding to firmware to prototyping using ESP32 control firmware (C++), breadboard with driver electronics, and 3D printed enclosure (SolidWorks)

Robot Interpretation of Human Speech | Research Group, University of Oxford, UK **Oct 2025 – Present**

- Submitted human robot interaction research to MIT journal (TACL) which publishes papers in computational linguistics and NLP
- Evaluated whether household robots correctly execute human instructions when the same command is phrased in different ways
- Performed data validation and statistical analysis of task success rates across 4,670 robot instructions, with 17 distinct framing types used (1 neutral baseline + 16 variations).

ENGINEERING TEAMS

UCSD Rocket Propulsion Laboratory | *Rocket Engineer* **Oct 2025 – Present**

Student team designing, building, and flying solid propellant rockets for the Spaceport America Cup competition.

- Designing a 9.4 ft two stage demonstrator rocket with a Marman clamp staging mechanism actuated by melting a Nichrome wire, incorporating feedback from SpaceX Engineers. Scheduled to launch in September 2026 to achieve stage separation and full recovery of both stages and avionics. Serves as a prototype for a 15 ft rocket candy vehicle (35% sorbitol / 65% potassium nitrate).
- Launched a 24.21 in long, 1.6 in diameter G-class rocket at the Friends of Amateur Rocketry facility, reaching 900 ft/s. Simulated and programmed using OpenRocket and MATLAB

UCSD Triton Robotics | *Mechanical Engineer* **Jan 2025 – Oct 2025**

Student team building paintball shooting robots for the international RoboMaster competition

- Redesigned 42 mm projectile launcher assembly to increase stability and speed of the robot through a lower center of gravity.
- Redesigned the mecanum wheel suspension on a new chassis to improve traction and stability over uneven terrain.

PROFESSIONAL EXPERIENCE

Bayer, Vividion Therapeutics | *Data Engineer Intern* **Jun 2026 – Aug 2026**

Biopharma company running cancer drug trials and discovery programs globally.

- Developed Python based validations checks to automatically review and classify 12,000+ lines of patient data. Eliminated manual review and saved doctors 10+ hours each month.
- Partnered with clinical research, operations, formulation scientists, and cross functional teams supporting an 88 patient colorectal cancer trial across 63 global sites. Studied the end to end drug discovery and trial process.

Merantis | *AI Engineer Intern* **Jun 2024 – Sep 2024**

AI/ML startup automating regulatory compliance documentation for medical devices

- Built a multimodal model to classify medical device images and retrieve matching compliance requirements from regulatory documents. Used an attention based architecture to align image and text.
- Led a 3 person team implementing convolution, activation, and cross entropy loss functions to train a CNN classifier from scratch.